UDESC UNIVERSIDADE DO ESTADO DE SANTA CATARINA

Centro de Ciências Tecnológicas - CCT - Joinville
Departamento de Matemática

Lista 3 - Geometria Analítica
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1. Transforme os pontos dados em coordenadas cartesianas para coordenadas polares, de duas maneiras diferentes e represente-os graficamente.

   (a) $A(1,1)$
   (b) $B(2,-2)$
   (c) $C(\sqrt{3},1)$
   (d) $D(4,0)$
   (e) $E(0,-3)$

2. Transforme os pontos dados em coordenadas polares para coordenadas cartesianas, representando-os graficamente.

   (a) $A\left(1,\frac{\pi}{2}\right)$
   (b) $B\left(-2,\frac{49\pi}{6}\right)$
   (c) $C\left(3,\frac{-5\pi}{3}\right)$
   (d) $D\left(0,\frac{\pi}{9}\right)$
   (e) $E(7,\pi)$

3. Identifique cada uma das curvas abaixo, represente-as geometricamente e converta as equações para coordenadas polares.

   (a) $x^2+y^2=4$
   (b) $(x-2)^2+y^2=4$
   (c) $y=\sqrt{3}x$
   (d) $x+2y=4$
   (e) $x^2+(y+1)^2=3$
   (f) $y=x^2$
   (g) $\frac{x^2}{4}+\frac{y^2}{9}=1$
   (h) $x^2-y^2=1$

4. Determine as equações cartesianas das curvas abaixo dadas em coordenadas polares e represente geometricamente.

   (a) $r=2$
   (b) $r=2\cos\theta$
   (c) $r=2\sin\theta$
   (d) $r\cos\left(\theta+\frac{\pi}{4}\right)=\sqrt{2}$
   (e) $r\cos\theta=3$
   (f) $r=\dfrac{2}{3\sin\theta-5\cos\theta}$
   (g) $r=1-2\sin\theta$
   (h) $r=1+\frac{1}{2}\cos\theta$

5. Identifique as curvas a seguir e represente-as geometricamente usando o sistema de coordenadas polares (ou seja, um círculo polar).

   (a) $r=4\sin\theta$
   (b) $r=3\cos(2\theta)$
   (c) $r=\sqrt{-3\sin(2\theta)}$
   (d) $r=2-2\cos\theta$
   (e) $r=2\sin(5\theta)$
   (f) $r=2\sin(2\theta)$
   (g) $r=\sqrt{\cos(2\theta)}$
   (h) $r=3+2\sin\theta$
   (i) $r=\theta$
   (j) $r\theta=1$

6. Identifique as curvas abaixo, desenhe a região R do plano simultaneamente interior as curvas e determine os pontos de interseção.

   (a) $r=3$ e $r=3\cos(2\theta)$
   (b) $r=2+2\sin\theta$ e $r=2$
   (c) $r=\cos(3\theta)$ e $r=\sin(3\theta)$
   (d) $r=\sqrt{\cos(2\theta)}$ e $r=\sqrt{\sin(2\theta)}$;
   (e) $r=2\left(1+\sin\theta\right)$ e $r=2\left(1+\cos\theta\right)$.

7. Nas Figuras 1 e 2 estão representadas algumas das curvas dadas em coordenadas polares pelas equações:
   - $C_1:\ r=2$;
   - $C_2:\ r=2+2\cos\theta$;
   - $C_3:\ r=-4\cos\theta$;
   - $C_4:\ r=2\sin(3\theta)$;
   - $C_5:\ r=2\sin\theta$;
   - $C_6:\ r=\cos(3\theta)$;
   - $C_7:\ r=2-\sin\theta$
   - $C_8: r=1-2\sin\theta$.

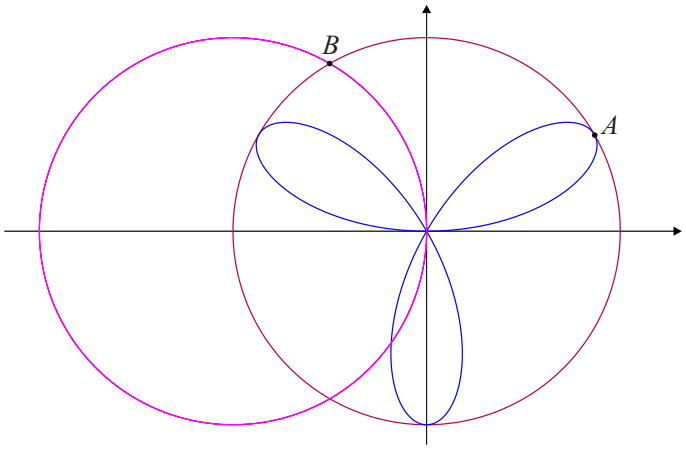


Figura 1: (a)

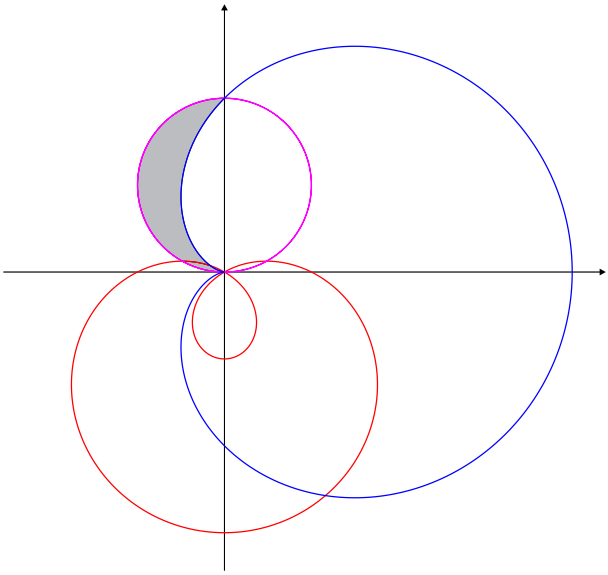


Figura 2: (b)

(a) Determine as curvas que estão representadas na Figura 1 e os pontos A e B em coordenadas polares e cartesianas.

(b) Determine as curvas que estão representadas na Figura 2 e descreva a região hachurada.

Respostas:

1. .

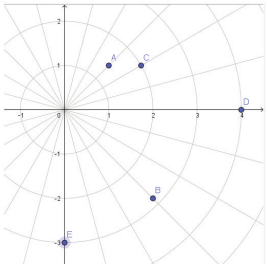
   (a) $A\left(\sqrt{2}, \frac{\pi}{4}\right)$

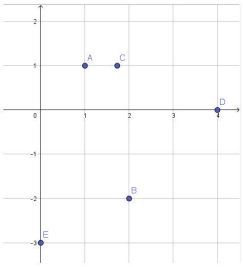
   (b) $B\left(2\sqrt{2}, \frac{7\pi}{4}\right)$

   (c) $C\left(2, \frac{\pi}{6}\right)$

   (d) $D(4, 0)$

   (e) $E\left(3, \frac{3\pi}{2}\right)$





2. .

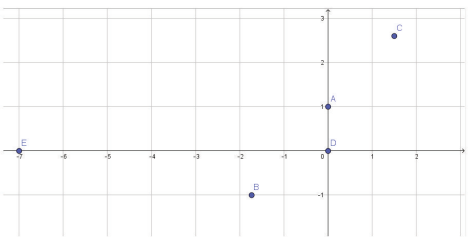
   (a) $A(0, 1)$

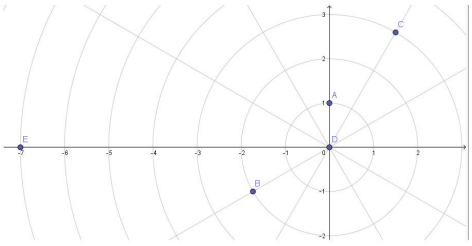
   (b) $B(-\sqrt{3}, -1)$

   (c) $C\left(\frac{3}{2}, \frac{3\sqrt{3}}{2}\right)$

   (d) $D(0, 0)$

   (e) $E(-7, 0)$





3. .

   (a) $r = 2$

   (b) $r = 4\cos(\theta)$

   (c) $\theta = \frac{\pi}{3}$

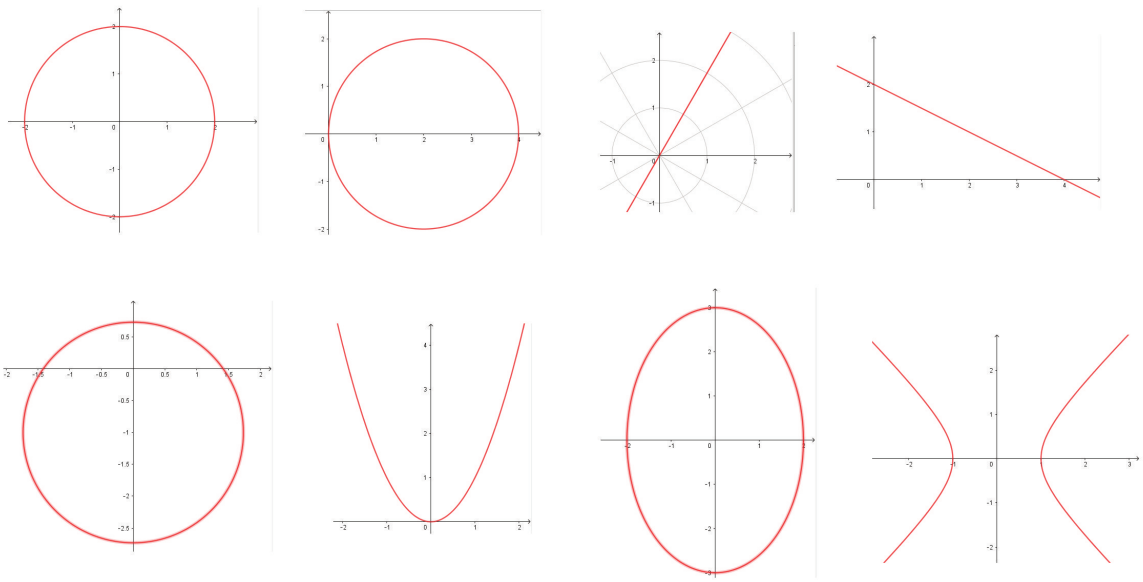
   (d) $r(\cos\theta + 2\sin\theta) = 4$

   (e) $r^2 + 2r\sin\theta - 2 = 0$

   (f) $r = \tan(\theta)\sec(\theta)$

   (g) $r^2(4 + 5\cos^2\theta) = 36$

   (h) $r^2\cos(2\theta) = 1$



4. .

(a) $x^2 + y^2 = 4$

(b) $x^2 + y^2 = 2x \quad \Leftrightarrow \quad (x-1)^2 + y^2 = 1$

(c) $x^2 + y^2 = 2y \quad \Leftrightarrow \quad x^2 + (y-1)^2 = 1$

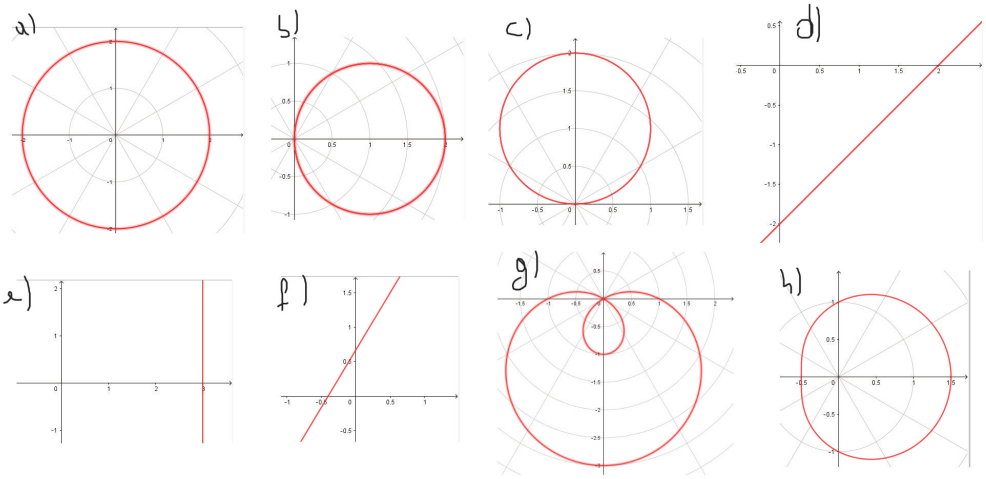
(d) $x - y = 2$

(e) $x = 3$

(f) $3y - 5x = 2$

(g) Limaçon com laço; $x^2 + y^2 = \sqrt{x^2 + y^2} - 2y$

(h) Limaçon sem laço; $x^2 + y^2 = \sqrt{x^2 + y^2} + \dfrac{x}{2}$



5. .

(a) Circunferência $C(0, 2)$ e raio 2.

(b) Rosácea de 4 pétalas.

(c) Cardioide (com simetria na reta $\theta = \frac{\pi}{2}$).

(d) Cardioide (com simetria no eixo polar).

(e) Rosácea de 3 pétalas.

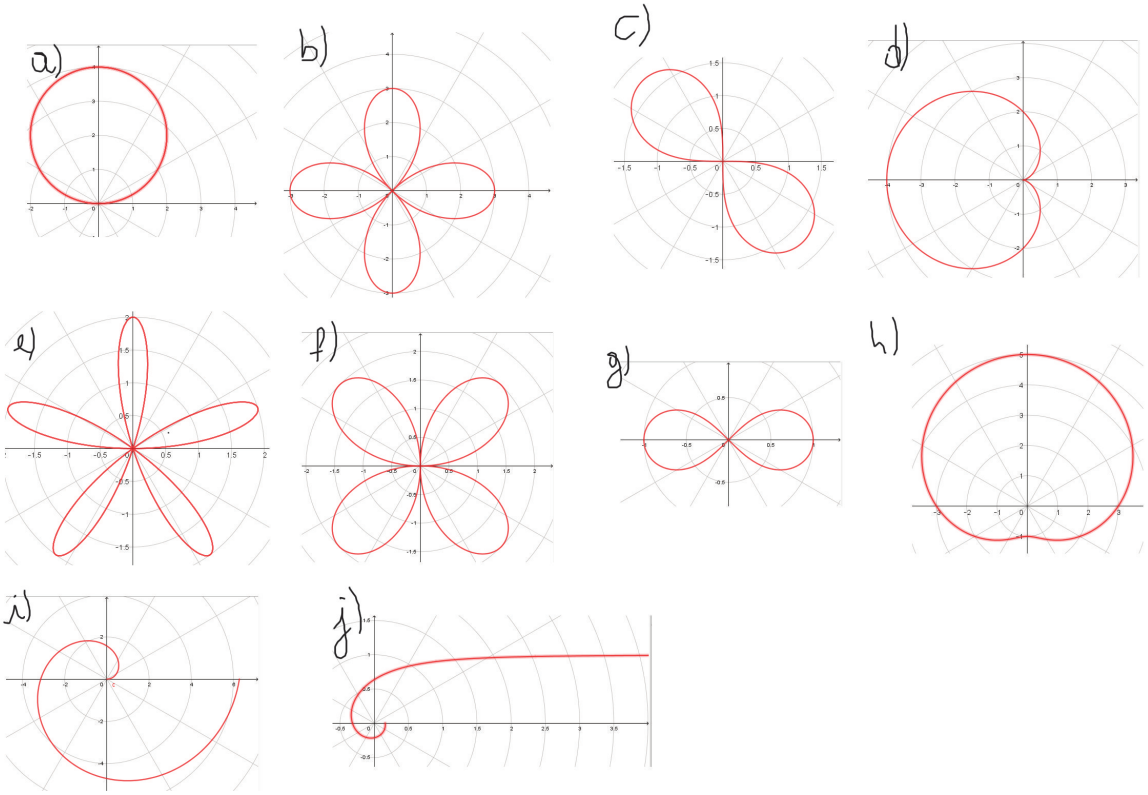
(f) Rosácea de 4 pétalas.

(g) Lemniscata.

(h) Cardioide (com simetria na reta $\theta = \frac{\pi}{2}$).

(i) Espiral.

(j) Espiral.



6. .

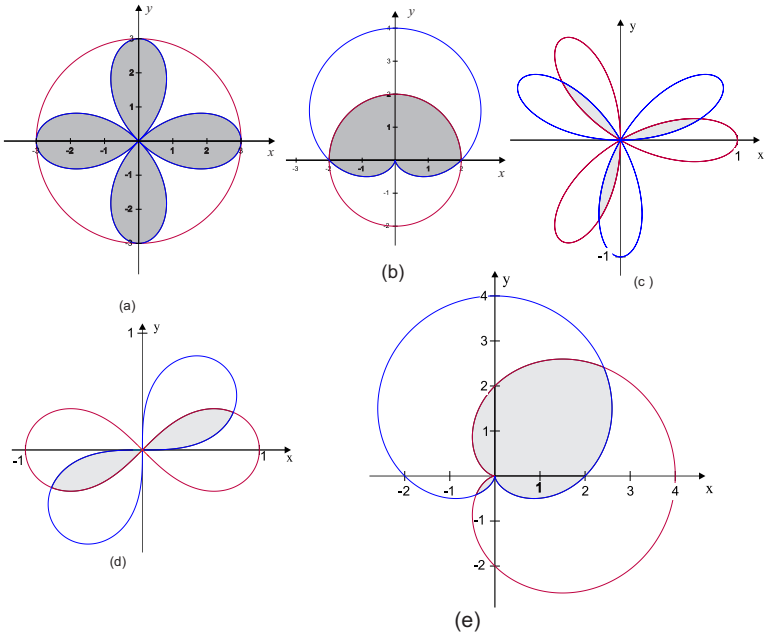
(a) Uma circunferência e uma rosácea de 4 pétalas; $(3,0)$, $(3,\pi)$, $(3,\pi/2)$, $(3,3\pi/2)$

(b) Um cardioide e uma circunferência; $(2,0)$ e $(2,\pi)$

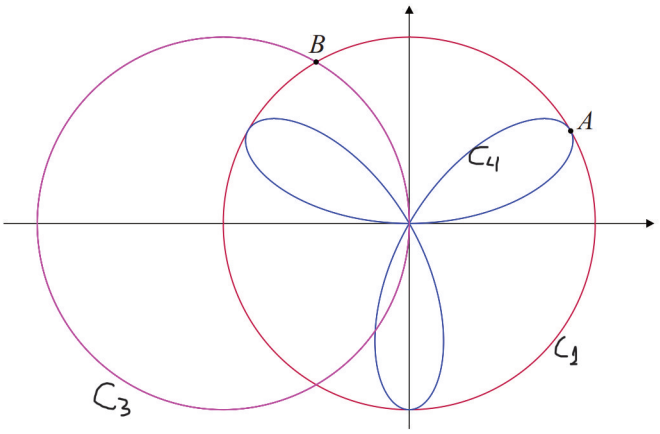
(c) Duas rosáceas de 3 pétalas; $(\sqrt{2}/2,\pi/12)$, $(-\sqrt{2}/2,5\pi/12)$ e $(\sqrt{2}/2,3\pi/4)$

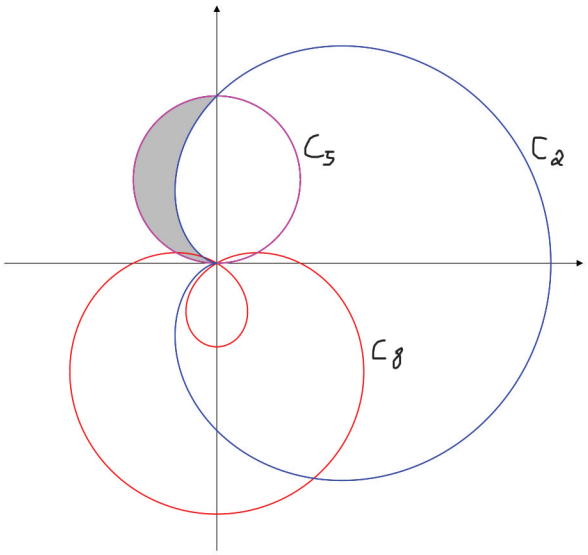
(d) Duas lemniscatas; $(\sqrt{2}/2,\pi/8)$ e $(\sqrt{2}/2,9\pi/8)$

(e) 2 cardioides; $(2+\sqrt{2},\pi/4)$ e $(2-\sqrt{2},5\pi/4)$



7. .





(a) $A(2, \frac{\pi}{6})$ e $A(\sqrt{3}, 1)$; $B(2, \frac{2\pi}{3})$ e $B(-1, \sqrt{3})$.

(b) $R = \{(r, \theta) : \ \frac{\pi}{2} \leq \theta \leq \pi, \ 2 + 2\cos(\theta) \leq r \leq 2\sin(\theta)\}$.